

```
(including wake-boot from deepsleep)
#import esp
#esp.osdebug(None)
import gc
import webrepl
webrepl.start()
gc.collect()
```

```
iAyan:~ iAyan$ ampy --port /dev/tty.
SLAB_USBtoUART put main.py
iAyan:~ iAyan$ ampy --port /dev/tty.
SLAB_USBtoUART ls
boot.py
webrepl_cfg.py
main.py
```

This is handy to transfer main.py file to board and will run as soon as the board boots after boot.py file.

Hardware tinkering with MicroPython

Hardware hacking - GPIO. Connect the positive side, that is, anode of the LED to pin 15 on ESP8266 board via a 330-ohm resistor and the negative side or cathode (shorter leg) to

the ground pin.

```
import machine
import time

pin_number = 15
pin = machine.Pin(pin_number, machine.
Pin.OUT)

while True:
    pin.high()
    time.sleep(1)
    pin.low()
    time.sleep(1)
```

Networking with MicroPython

Networking is a powerful feature of microcontrollers, thanks to the Internet of Things. ESP8266 is a low-cost Wi-Fi development board and, because of MicroPython, it exposes the required APIs to access network- ing on ESP8266.

A script to connect your board to your router and open the gate to the Internet is:

```
>>> import network
```



Fig. 6: Connecting to ESP8266

```
>>> AP_NAME = " "
>>> PASSWORD = " "
>>> ap = network.WLAN(network.STA_IF)
>>> ap.active(True)
#5 ets_task(4020ed88, 28, 3fff9708, 10)
>>> ap.connect(AP_NAME, PASSWORD)
>>> ap.ifconfig()
('192.168.1.127', '255.255.255.0',
'192.168.1.1', '192.168.1.1')
```

Congratulations! ESP8266 is now a device on your local network with just five lines of Python code. Now you can address WebREPL over this IP address. **EFY**

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Application: LED Lighting Devices



[Development background]

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